

A 10.2-to-14.82-GHz Low-Phase-Noise All-Digital Phase-Locked Loop

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ABSTRACT

This paper presents a subsampling bang-bang all-digital phase-locked loop (SS-BB-ADPLL) implemented in the TSMC 40nm process. The proposed PLL architecture replaces conventional time-to-digital converter (TDC) with a bang-bang phase detector (BBPD), effectively reducing circuit complexity. It also utilizes a transformer-feedback dual-core digitally controlled oscillator, which further reduces phase noise and achieves low jitter performance. The design achieves a measured root mean square (RMS) jitter of 196.8 fs in integer-N mode and 707.7 fs in fractional-N mode, while demonstrating a frequency tuning range from 10.2 GHz to 14.82 GHz.

Keywords—Oscillator, Phase-Locked Loop (PLL), low jitter, phase noise, Time to Digital Converter (TDC), bang-bang phase detector (BBPD)

INTRODUCTION

With the continuous development of communication technologies, wireless transceivers and RF systems on-chip have imposed highly challenging requirements on the jitter of PLL. Therefore, reducing the jitter of PLLs has always been a hot topic. Analog PLLs, after decades of development, are now relatively mature. However, since Professor Robert Bogdan Staszewski published the first paper on all-digital PLLs (ADPLLs) in 2004 [1], many studies have been dedicated to this topic. Compared to analog PLLs, digital PLLs offer unique advantages, such as shorter design cycles, strong portability, and reduced susceptibility to PVT (process, voltage, temperature) variations.

Currently, there are two mainstream architectures for all-digital PLLs: one based on dividers [2] and the other based on counters [1]. This paper adopts the divider-based architecture, which, due to its subsampling nature, employs a fixed division ratio instead of a multi-modulus division. In this article, we will introduce the design processes of various critical modules within the PLL.

ADPLL DESIGN

System Design

The overall architecture of the SS-BB-ADPLL proposed in this paper is illustrated in Fig. 1. Its core operational workflow is summarized as follows. First, the reference clock undergoes two-stage digital-to-time converter DTC delay processing before being fed into the BBPD. Then, the phase error signals generated by the BBPD are processed through a proportional-integral (PI) digital loop filter (DLF), whose output modulates dual digital-to-analog converters (DACs) for hierarchical oscillator frequency adjustment. The first-level DAC achieves a broader frequency tuning, while the second-level DAC achieves a narrower frequency tuning. Finally, the output of the oscillator is divided by four and fed back to the BBPD, completing the main loop operation of the PLL presented in this paper. In addition to the main loop, the PLL proposed in this paper also includes Automatic Frequency Control (AFC) and Gear Shifting (GS). AFC assists the subsampling PLL in achieving lock, while GS is used to adjust the lock time and the RMS jitter after locking.

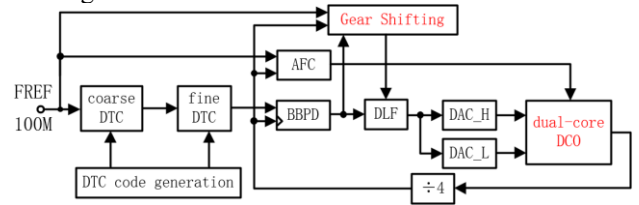


Fig. 1 The architecture of the proposed SS-BB-ADPLL

Oscillator design

The oscillator in the PLL features a transformer-feedback structure [3] and has been extended to a dual-core configuration, as shown in Fig. 2.

In this configuration, C1 and C2 are the coarse and medium tuning digital capacitor arrays placed at the drains of M1 and M2, while C3 and C4 are varactors located at the sources of M1 and M2. The inductive network comprises a primary inductor ($L_p = 300$ pH), a secondary inductor ($L_s = 220$ pH), and a mutual coupling coefficient $k = 0.41$. The width-to-length ratio (W/L) of M1 and M2 is $20\mu\text{m}/40\text{nm}$. This structure, through the coupling of the drain and source inductors, can achieve a higher output voltage swing, thereby reducing phase noise. Moreover, placing the varactors at the sources rather than the drains also aids in achieving finer frequency tuning.

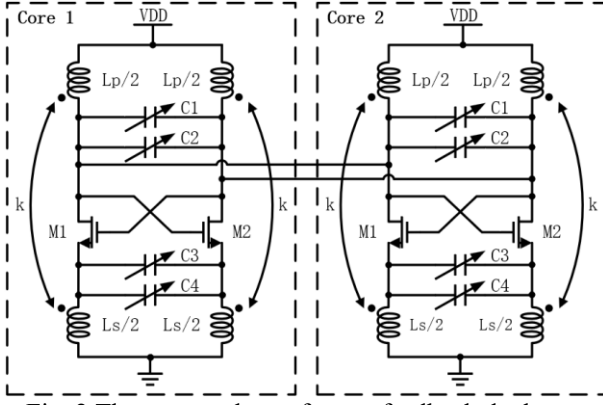


Fig. 2 The proposed transformer-feedback dual-core DCO structure

Gear Shifting design

The proportional gain (k_p) and integral gain (k_i) of the PI filter affect the loop bandwidth and phase margin of the PLL. A larger loop bandwidth in the PLL results in a shorter lock time, but a poorer RMS jitter after locking; whereas a smaller loop bandwidth leads to a longer lock time, but a better RMS jitter after locking. Therefore, to alleviate the contradiction between lock time and phase noise, a GS module is introduced. In the initial locking phase, the loop bandwidth of the PLL is increased to speed up locking. Once locking is complete, the GS module will adjust the values of k_p and k_i to reduce the loop bandwidth. The system will then re-lock and continue to adjust the values of k_p and k_i until the jitter performance is optimal.

However, determining whether the phase-locked loop has completed locking is a key issue. Fig. 3 illustrates the architecture of the GS module in the proposed design. The BBPD with a dead zone, which is used to assist in determining whether the loop is locked, consists of a delay chain, D flip-flops, and an encoder. Fig. 3(bottom right) depicts the correlation between the input time difference Δt and output response. Additionally, the size of the dead zone (denoted as T_{dz}) can be adjusted by adjusting the delay chain. The process for determining whether the loop is locked is as follows: within a specific period, if BBPD_DZ_out remains zero and the average value of BBPD_out is close to zero, then the loop is determined to be locked, and subsequently, the k_p and k_i of the loop filter are adjusted.

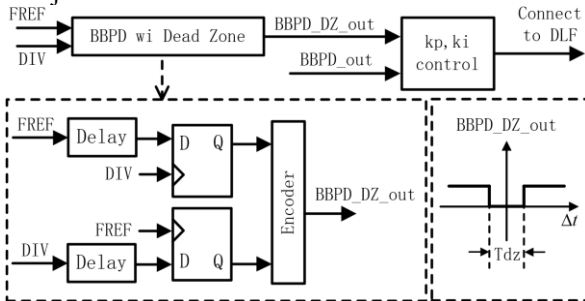


Fig. 3 The architecture of the proposed GS module

MEASUREMENT RESULT

Fig. 4 illustrates the chip micrograph. The chip is fabricated using TSMC 40nm technology and features a core area of 0.71 mm^2 .

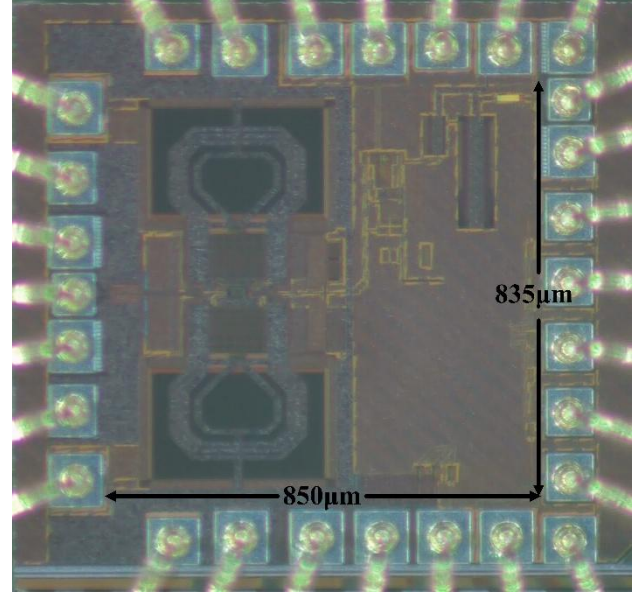


Fig. 4 Chip photo of the proposed ADPLL



Fig. 5 Test results during integer locking



Fig. 6 Test results during fractional locking

The measurement setup utilizes a 100 MHz crystal oscillator as the reference clock input. As the oscillator's

output is measured after being divided by two, the DCO output frequency is expected to be twice the test frequency. Fig. 5 presents the test results when the PLL is in integer lock, showing a DCO output frequency of 12 GHz and an RMS jitter of approximately 196.8 fs. Fig. 6 presents the test results when the PLL is in fractional lock, showing a DCO output frequency of 12.00272 GHz and an RMS jitter of approximately 707.7 fs.

A comparison of the proposed PLL with other PLLs is presented in Table 1. It can be seen from Table 1 that the PLL designed in this paper has a wider frequency range and a relatively better FoMT value.

TABLE 1
PERFORMANCE SUMMARY AND COMPARISON

	[4]	[5]	[6]	This work
Technology	40nm	90nm	28nm	40nm
Architecture	BB-DPLL	DPLL	BB-DPLL	SS-BB-ADPLL
Type	Int.N	Frac.N	Frac.N	Frac.N
Frequency (GHz)	21.4-25.1	9.2-12	15.6-18.2	10.2-14.82
FTR(%)	15.9	26.4	15.4	36.9
Reference (MHz)	390	40	200	100
DC Power (mW)	64	50	-	40
Jitter(fs)	394	597	-	197(int) 708(frac)
¹ FoM(dB)	-230 (int)	-227 (frac)	-237 (frac)	-238(int) -227(frac)
² FoMT(dB)	-234 (int)	-236 (frac)	-241 (frac)	-249(int) -238(frac)

$$^1FoM = 20\log_{10}\left(\frac{Jitter}{1s}\right) + 10\log_{10}\left(\frac{P_{DC}}{1mW}\right)$$

$$^2FoMT = FoM - 20\log_{10}\left(\frac{FTR}{10\%}\right)$$

CONCLUSION

This paper presents a subsampling bang-bang all-digital phase-locked loop implemented in the TSMC 40nm process. The SS-BB-ADPLL features a dual-core oscillator with transformer feedback and incorporates a GS module, achieving a frequency range of 10.2 to 14.82 GHz. The RMS jitter is 196.8 fs and 707.7 fs under integer-N and fractional-N locking, respectively, with corresponding FoMT values of -249 and -238.

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